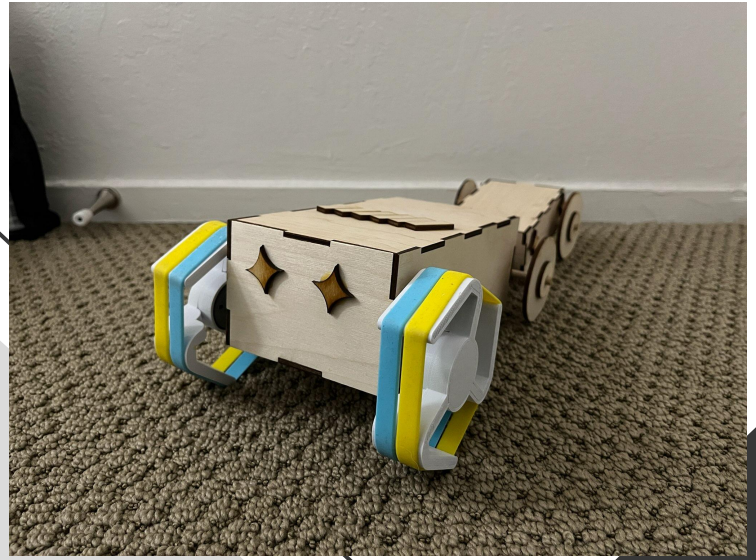


Insecure

Final Project: Med/High-Fidelity Bluetooth Controlled Car

Des Inv 22: Fall 2022

12/9/2022

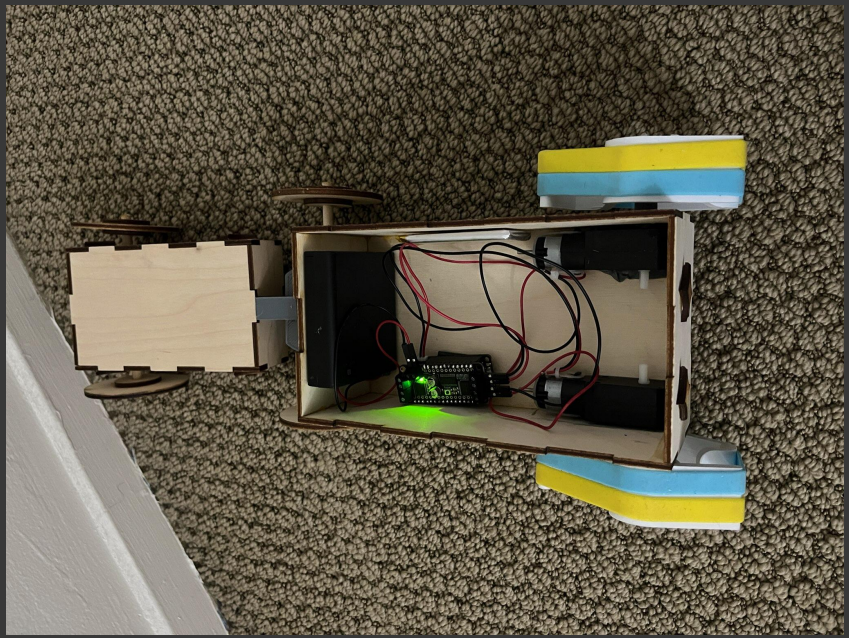
Christy Quang and Sophia Matthews

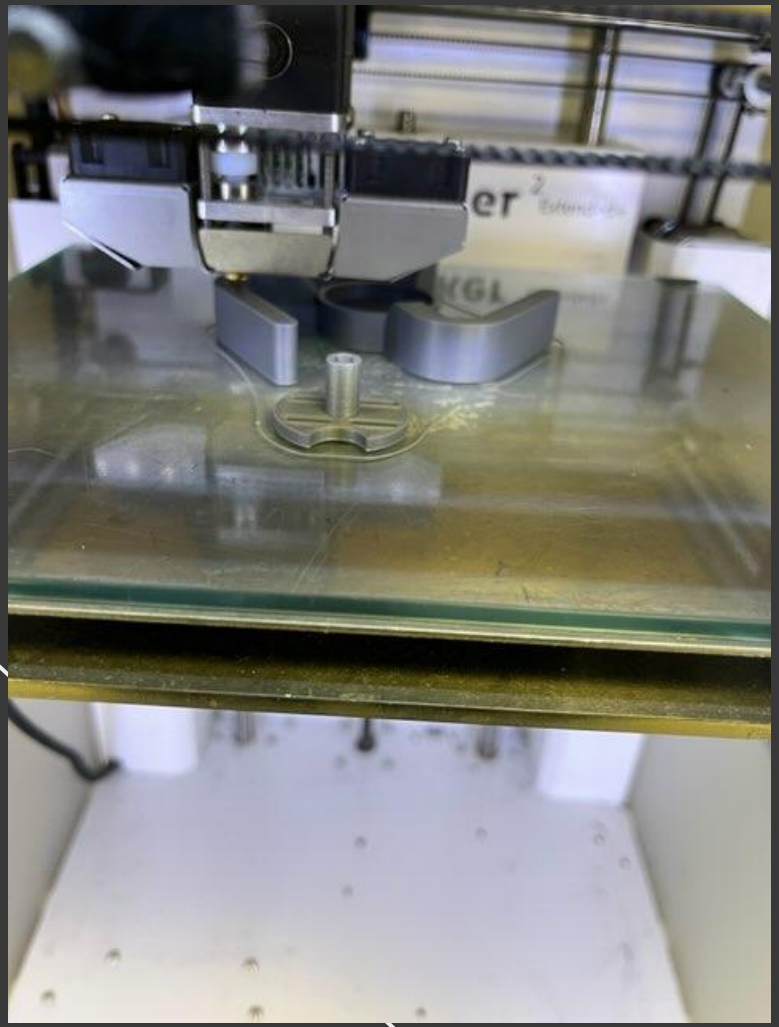
DESCRIPTION

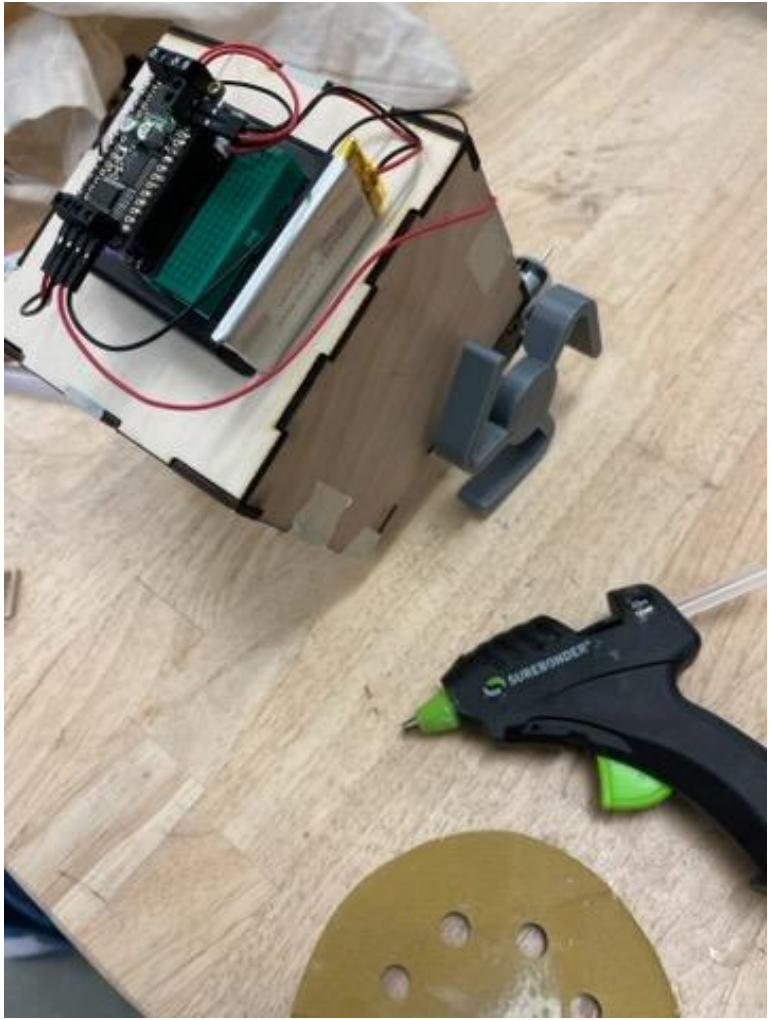
A bluetooth controlled car with two front driver wheels capable of powering through the different obstacles within the course.

Video Link: https://youtu.be/zQaoZfZG2LU

Unfortunately, we weren't able to get the course walkthrough since Sophia wasn't at the showcase, but we were able to finish the course and get the balancing extra credit after using one reset.



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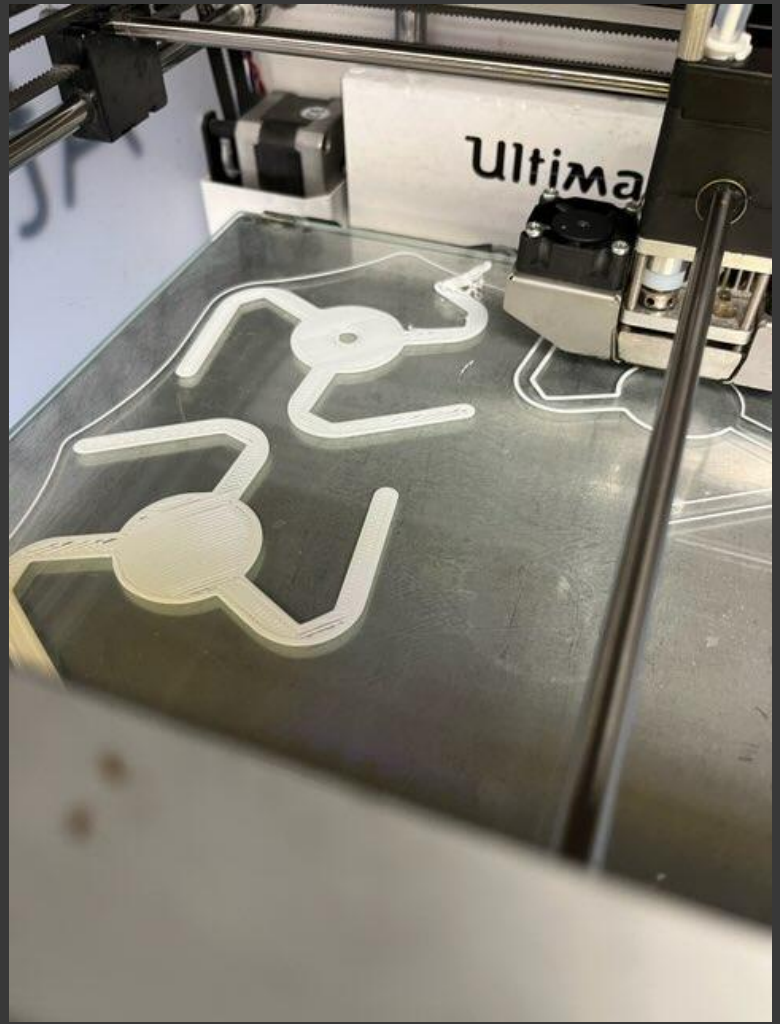
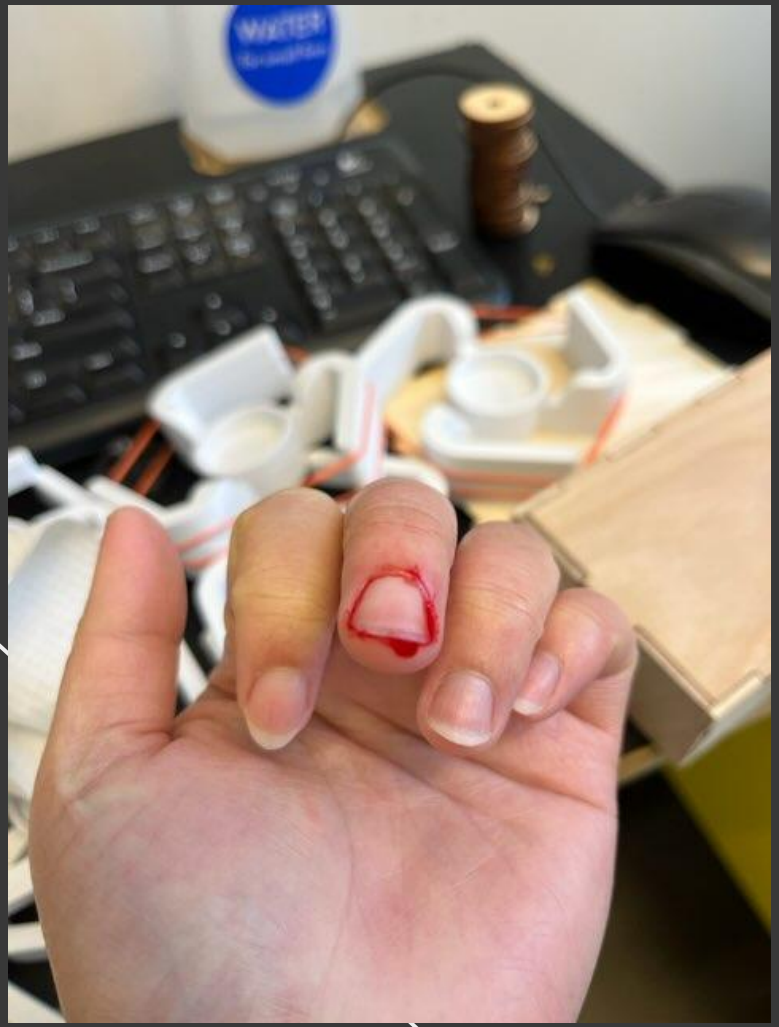
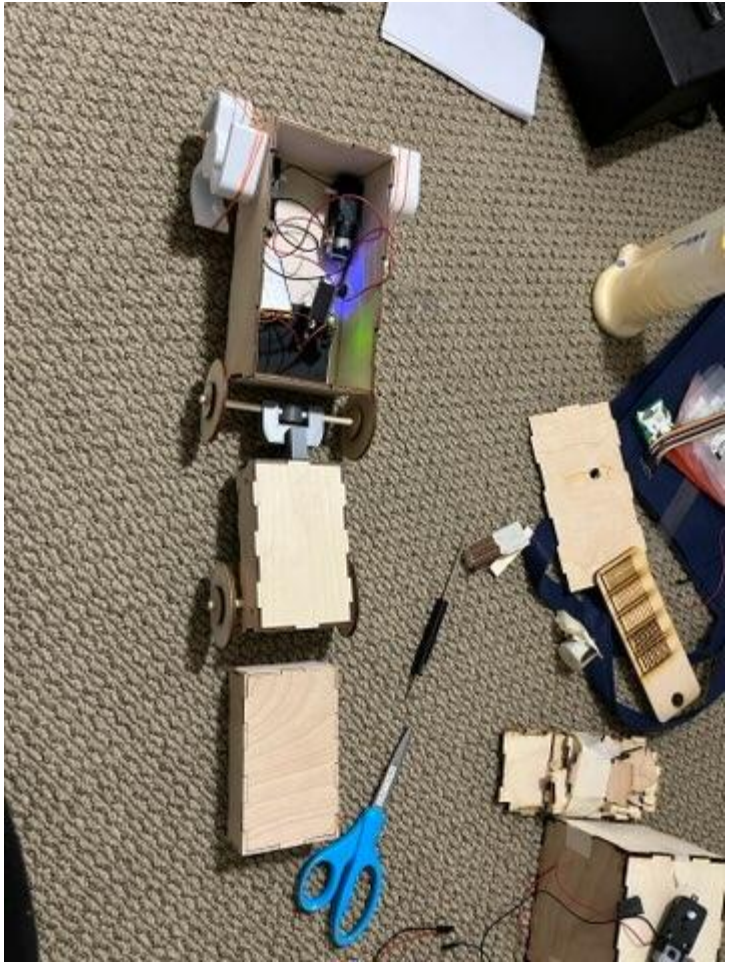
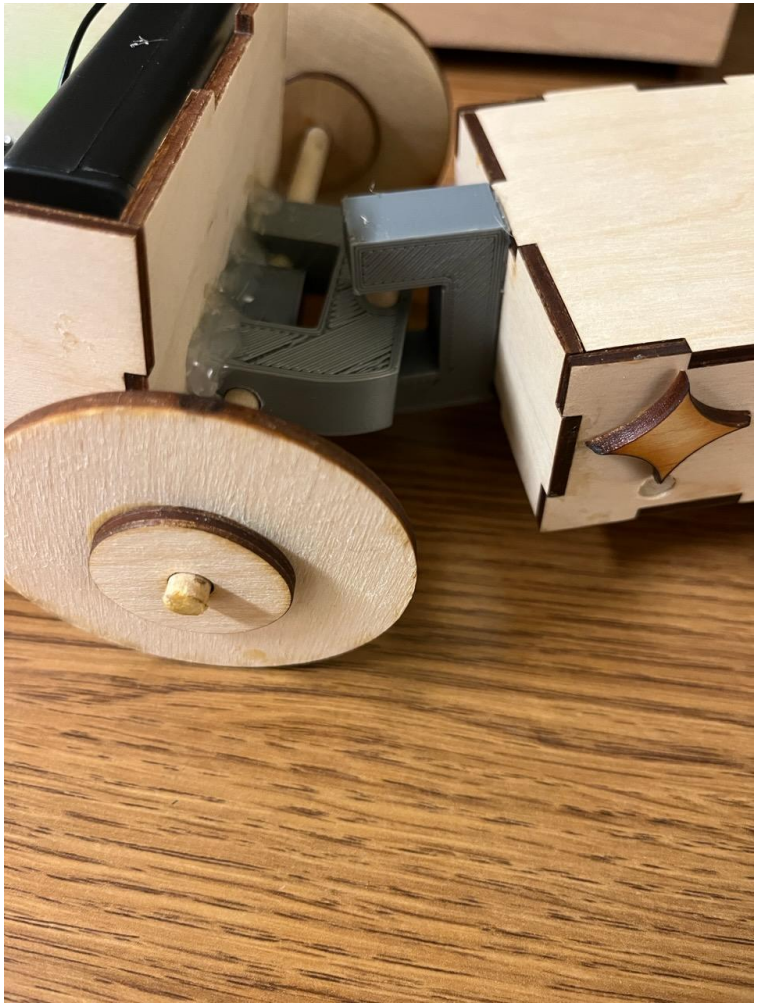
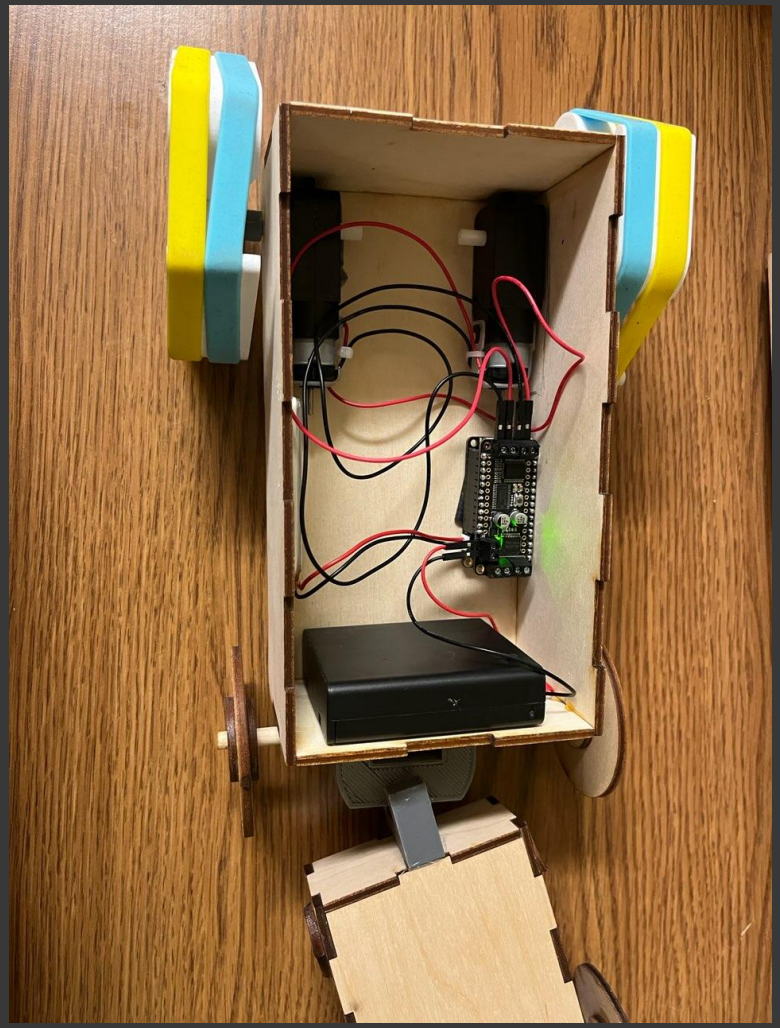

Ultima


WATER







Background

Experience

- I have limited background experience with electronics and bluetooth powered items. There was one lab in EECS16A that involved bluetooth control but I ended up dropping the class so I don't remember anything about that.

Design Choices

- The basic structure of a car is to have a body and some wheels. The bulk of this project is designing the wheels and the body of the car to be able to support it through the different obstacles. We chose to have our wheels imitate the circular shape of a normal wheel and to have a small body since we wanted our car to recreate a snake structure.

Improvements

- The foundation of the wheels is extremely unstable since the gear motors aren't attached to the outside of the car. As a result, the dowel holding the wheel is shorter, causing the wheels to be more unstable. Cutting a hole out for the motors would increase stability.

PROCESS

Debugging the Arduino was a long part of the project since the board given to the class made the bluetooth connection unable to work.

The first part of the project was designing the wheels since testing is unable to occur without the wheels being attached to the car. Getting the wheels printed was a struggle since there was high demand for the 3D printers and each print takes a bit of time.

After printing out different iterations of the wheel with improvements over a couple of days, the wheels could be attached to the wheel hub and gear motor. After printing out a makeshift laser cut box for testing, more improvements were made regarding the size and body of the car.

In order to replicate the snake-like structure, universal joints were 3D-printed in order to connect the different bodies to each other. Additional wheels that aren't attached to gear motors were printed in order to help the snake's body move. To increase stability and grip of the wheels, rubber bands were wrapped around the front two 3D printed wheels. This significantly improved the car's movement across the floor.

Bill of Materials

- 1 - 1/8” Wooden dowel
- 1 - Adafruit Feather 32U4 Bluefruit
- 2 - Right Angle Gear Motor
- 1 - Battery box
- 4 - Double AA batteries
- 4 - Rubber bands
- 1 - Microcontroller
- 1 - USB Cable
- 1 - Portable battery
- Birch plywood
- Wood glue
- Handsaw
- Sandpaper
- Paste wax
- Laser cutter
- X-acto knife
- Black Gaffers Tape
- 3D filament
- Double-sided tape

CODE

After switching the boards, I was able to get the co
to move using the same code that Berk had
provided to the class.

Example Code —>

Ada_BLE_RC | Arduino 1.8.19

Verify

Ada_BLE_RC | BluefruitConfig.h | packetParser.cpp

```
#include <string.h>
#include <Arduino.h>
#include <SPI.h>


#include "Adafruit_BLE.h"
#include "Adafruit_BluefruitLE_SPI.h"
#include "Adafruit_BluefruitLE_UART.h"

#include "BluefruitConfig.h"

#include <Wire.h>
#include <Adafruit_MotorShield.h>
// #include "utility/Adafruit_PWMServoDriver.h"
// #include <Servo.h>

// Create the motor shield object with the default I2C address
Adafruit_MotorShield AFMS = Adafruit_MotorShield();

// And connect 2 DC motors to port M3 & M4 !
Adafruit_DCMotor *L_MOTOR = AFMS.getMotor(4);
Adafruit_DCMotor *R_MOTOR = AFMS.getMotor(3);

//not used, testing acceleration
// int accelTime = 200;

//Name your RC here
String BROADCAST_NAME = "adafruit red robot rover";
```

1 Adafruit Feather M0 (SAMD21), Small (-Os) (standard), Arduino, Off on /dev/cu.usbmodem14201

Reflection

Surprises Encountered:

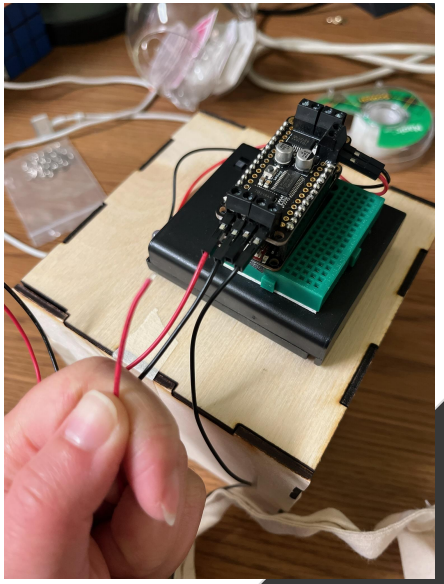
3D printing the wheels was surprisingly difficult since each print would take a couple of hours and since the 3D printers were in such high demand, it was difficult to get the print started. Additionally, every time a design choice was changed, this meant that the wheel had to be reprinted. Getting the car to be structurally stable was another unanticipated obstacle. We had glued our box together but didn't realize that the gear dowels would be shorter, thus making the wheels less structurally sound.

Reflection

Frustrating Parts:

One of the frustrating parts was experimenting with wheel design. Without being able to actually glue down the wheels (this meant that we would have to reprint), this meant that our testing was based off of inference. After solidifying our car and wheel design, we didn't realize that the wheel would be unstable so that was another unanticipated obstacle. Securing the universal joint between the two car bodies was also frustrating since adhesive tape, wood glue and hot glue wouldn't secure the 3D filament with the wood.

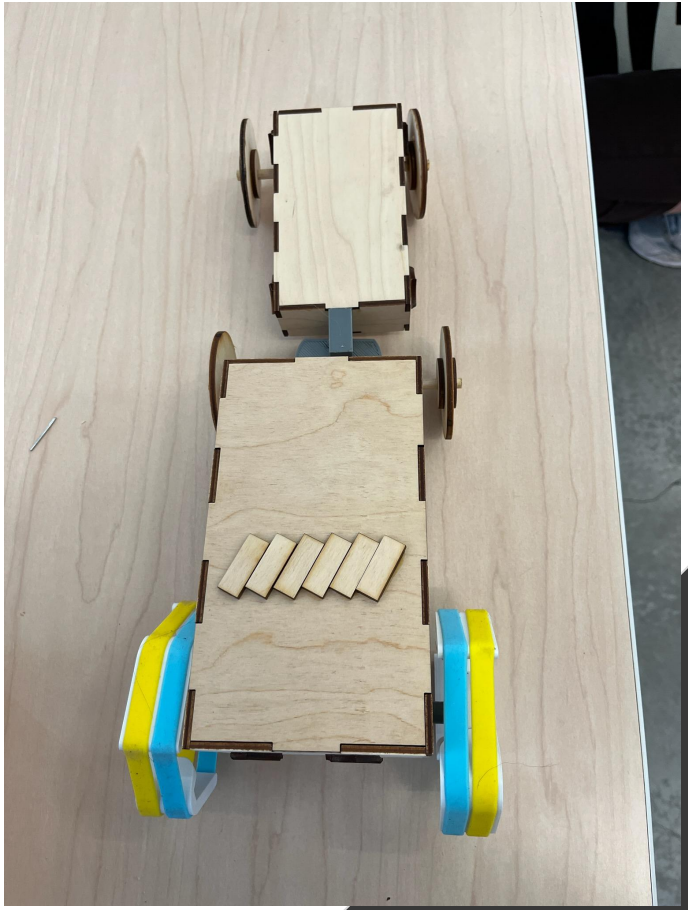
Additionally, the wires for the gear motors were flimsy so two of the motors used had to be replaced.



Reflection

Different Approaches:

I would redesign the secondary wheels to be a tiny bit thicker to allow for the car to roll easier. Prior to gluing the box together, I would cut a small rectangle for the gear motors to be put flesh against to, allowing for the wheels to be more secure. To cover up the hole in the car body, laser cut plywood would be placed over that. Finally, I would make the universal joint more secure since after a bit of movement, the joint falls apart.



Bibliography

Jamesbruton, director. *YouTube*, YouTube, 24 Jan. 2022, https://www.youtube.com/watch?v=eUUanZVFN84. Accessed 30 Nov. 2022.

THANK YOU!